

Lecture 15: Number System Conversions (Part 2): Binary $\leftrightarrow$ Octal & Binary $\leftrightarrow$ Hexadecimal

Learning Objectives

By the end of this lecture, students will be able to:

- Explain why Octal and Hexadecimal are used in computing.
 - Convert Binary to Octal.
 - Convert Octal to Binary.
 - Convert Binary to Hexadecimal.
 - Convert Hexadecimal to Binary.
 - Convert Octal to Hexadecimal and vice versa using Binary.
 - Solve exam-style conversion problems quickly and accurately.
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1. Introduction

Teaching Script

First 30-Second SEO Phrase

Naturally include:

"In today's Computer Organization and Assembly Language lecture, we'll learn Binary to Octal, Octal to Binary, Binary to Hexadecimal, Hexadecimal to Binary, and Octal to Hexadecimal conversions using easy grouping methods with solved examples."

In the previous lecture, we learned how to convert numbers between Decimal and Binary.

Today, we'll continue our Number System Conversion chapter by learning Binary, Octal, and Hexadecimal conversions.

You might wonder: if computers understand only Binary, then why do we use Octal and Hexadecimal?

The answer is simple. Binary numbers become very long and difficult to read. Octal and Hexadecimal provide shorter, easier-to-read representations while still preserving the exact same value.

In today's lecture, we'll learn quick conversion techniques using simple grouping methods and solve many examples.

Let's begin.

2. Why Do We Use Octal and Hexadecimal?

Imagine writing a memory address like:

1111111110110101010101111110000

It's difficult to read.

The same value in Hexadecimal is:

FFDAABF0

Much shorter and easier.

Similarly:

77766525760

Octal is also shorter than Binary.

Advantages

- Easier to read.
 - Fewer digits.
 - Commonly used in programming, debugging, and memory addresses.
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3. Binary ↔ Octal Relationship

One Octal digit represents **3 binary bits** because:

$$2^3 = 8$$

So:

1 Octal Digit

↓

3 Binary Bits

4. Binary to Octal Conversion

Method

1. Start from the **rightmost bit**.
2. Group bits into **sets of 3**.
3. Add leading zeros if necessary.

4. Convert each group into one Octal digit.
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Octal Lookup Table

Binary	Octal
000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

Example 1

Convert:

101110_2

Group into 3 bits:

101

110

Convert:

$101 = 5$

$110 = 6$

Answer:

56_8

Example 2

Convert:

111001101_2

Groups:

111

001

101

Conversion:

7

1

5

Answer:

715₈

Example 3 (Padding Required)

Convert:

11011₂

Pad with leading zeros:

011

011

Convert:

3

3

Answer:

33₈

5. Octal to Binary Conversion

This conversion is even easier.

Replace every Octal digit with its **3-bit binary equivalent**.

Example 1

Convert:

537₈

Lookup:

5 → 101

3 → 011

$7 \rightarrow 111$

Answer:

101011111_2

Example 2

Convert:

12_8

Binary:

$1 \rightarrow 001$

$2 \rightarrow 010$

Result:

001010_2

You may remove unnecessary leading zeros:

1010_2

6. Binary ↔ Hexadecimal Relationship

One Hexadecimal digit represents **4 binary bits** because:

$$2^4 = 16$$

Therefore:

1 Hex Digit

↓

4 Binary Bits

7. Binary to Hexadecimal Conversion

Method

1. Start from the right.
 2. Group bits into **4-bit groups**.
 3. Add leading zeros if needed.
 4. Convert each group into one hexadecimal digit.
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Hexadecimal Lookup Table

Binary	Hex
--------	-----

0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A
1011	B

1100	C
1101	D
1110	E
1111	F

Example 1

Convert:

10101111_2

Groups:

1010

1111

Conversion:

A

F

Answer:

AF_{16}

Example 2

Convert:

11111111_2

Groups:

1111

1111

Answer:

FF_{16}

Example 3 (Padding Required)

Convert:

10111_2

Pad:

0001

0111

Convert:

1

7

Answer:

17_{16}

8. Hexadecimal to Binary Conversion

Replace every Hex digit with its **4-bit binary equivalent**.

Example 1

Convert:

$3A_{16}$

Binary:

$3 \rightarrow 0011$

$A \rightarrow 1010$

Answer:

00111010_2

Example 2

Convert:

$F2_{16}$

Binary:

$F \rightarrow 1111$

$2 \rightarrow 0010$

Answer:

11110010_2

9. Octal $\leftrightarrow$ Hexadecimal Conversion

There is **no direct conversion rule**.

Use Binary as the intermediate step.

Example

Convert:

57_8

Step 1: Octal $\rightarrow$ Binary

$5 \rightarrow 101$

$7 \rightarrow 111$

Binary:

101111_2

Pad to groups of 4:

0010

1111

Convert:

2

F

Answer:

$2F_{16}$

10. Conversion Summary

Conversion	Method
Decimal → Binary	Repeated Division
Binary → Decimal	Positional Weights
Binary → Octal	Group into 3 bits
Octal → Binary	Replace each digit with 3 bits
Binary → Hex	Group into 4 bits
Hex → Binary	Replace each digit with 4 bits
Octal ↔ Hex	Convert through Binary

11. Real-Life Analogy

Think of Binary as writing every word letter by letter.

Octal and Hexadecimal are like using abbreviations.

The **message stays the same**, but it's much easier to read.

12. Common Mistakes

Students often:

✗ Group Binary from the **left**.

✓ Always group from the **right**.

✗ Forget leading zeros.

✓ Add zeros on the **left** whenever required.

✗ Memorize only a few Hex values.

✓ Learn the complete lookup table from **0000** to **1111**.



Lecture Summary

Today, we completed the Number System Conversion chapter by learning Binary ↔ Octal and Binary ↔ Hexadecimal conversions.

We learned the grouping methods, solved multiple examples, and understood why Octal and Hexadecimal are widely used to represent Binary values more compactly.

We also covered Octal ↔ Hexadecimal conversion using Binary as the intermediate representation and discussed common mistakes that students should avoid in exams.

In the next lecture, we'll begin **Binary Arithmetic**, where we'll learn Binary Addition, Binary Subtraction, Binary Multiplication, and Binary Division. These operations are essential because the CPU performs all arithmetic internally in Binary.

Thank you for watching, and I'll see you in the next lecture.
